Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/04/2005
PI Name:	Pandya, Abhilash Ph.D.		
Project Title:	Real-Time Augmented Reality Development and Human Factors Assessment for the Special Purpose Dexterous Manipulator		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48202	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	07/07/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Baker, Kenneth (NASA/JSC) Maida, James (NASA/JSC) Ellis, Darin (Wayne State University) Chien, Chiun-Hong (NASA/JSC/HEI) Cao, Alex (Children's Hospital of Michigan) Nawab, Aditya (Wayne State University) Xu, Cheng-Zhong (Wayne State University) Auner, Greg (Wayne State University) Pace, John (NASA/JSC/LM) Bowen, Charles (NASA/JSC/LM) Sorensen, David (NASA/JSC/HEI) Chen, Wanli (Wayne State University) Woolford, Barbara (NASA/JSC-COTR)		
Grant/Contract No.:	NNJ04HI11A		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The major goal of this Rapid Technology Development grant is to provide the operator of the Special Purpose Dexterous Manipulator (SPDM) with simple, intuitive cues for control of the manipulator during grasping, removing, stowing and inserting orbital replacement units (ORUs) as opposed to just looking at multiple video views which requires extensive training and can be error prone. The science of presenting and displaying complex 3D images in an augmented and operationally meaningful way is the basis of this Space Human Factors Engineering research proposal. The technical development will be validated and proven with the appropriate Human Factors testing. Efficient methods of remote image data display will be implemented and tested against more traditional methods. The proposal's objective is to improve the human-robot interface to reduce operator work load, task time and to simplify operator training to make skill-based training practical as opposed to task-based training with particular focus on the SPDM and its training facility the DMT. Because of the risky environment of space, teleoperations are commonplace for NASA. Visualization of the remote data and its human performance impact are important to succesful teleoperations. During space missions, astronaut robotic operators primarily depend on visual feedback that is provided by strategically placed video cameras. Operating a robotic arm from indirect visual information poses many technical and human performance challenges for the operator. Astronauts performing remote functions like operating the SPDM or the RMS need to maintain mental 3D real-time environment maps for successful operations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The earth significance of this proposal is that successful completion of our first specific aim for this proposal (the core Augmented Reality Specifications) will provide an easily translatable kinematics-based AR system specifications for medical robotics. In addition, several important features in Medical Robotics systems (voice recognition and control, motion scaling and built-in structural intelligence) can also be of interest to NASA robotics. Medical robotics is an up-and-coming technology with tremendous potential and has some of the same visualization difficulties facing astronaut robotic operators. Medical Robotic systems like the Zeus (Computer Motions Inc.) and daVinci (Intuitive Inc.) (both presently being used in our collaborating sister facilities; Henry Ford Hospital—Zeus system and Children's Hospital—daVinci System both in Detroit, Michigan) are master-slave systems for minimally invasive surgery. Direct linkage of medical robotic systems to patient data and the optimal visualization of that data for the surgical team are of critical importance for successful operations.
Task Progress:	1. We have a very clearly defined project mandate and project plan that was agreed upon by all involved parties (including the customer at NASA/JSC). 2. Software and hardware constraints and system requirments for integration of a prototype AR system have been identified. 3. Software development for camera calibration and pose estimation has been completed and testing has begun at the DMT. 4. A testbed robotic system has been developed with features that will allow further advanced testing and technology development. 5 We have purchased and have experimented with an IBM A31P laptop which is currently on-board Space Station and is a candidate platform on which our final product could run. We have also installed the real-time Qnx operating system and have compiled and run some test code. 6. We have integrated an image processing technique (Retnex) developed at NASA Langley for live video processing. 7. We are also collaborating with a related sister project at the GRAF/NASA/JSC. This group is also running subject testing and has with our input developed an AR system for the DMT. This system differs from this projects AR in that it is a single end-effector mounted camera AR system. They have tested 2 subjects with their system and are sharing their test protocols and experience on the DMT with this project. We have provided extensive review of their subject testing protocol and our team members have been used as pilot test subjects for their tests.
Bibliography Type:	Description: (Last Updated: 06/30/2008)
Abstracts for Journals and Proceedings	Pandya A, Baker K, Maida J, Ellis RD, Chien C, Cao A, Nawab A, Xu CZ, Auner G, Pace J, Woolford B. "Real-Time Augmented Reality (AR) for Space Station Remote Manipulators." Bioastronautics Conference, NASA JSC, January 2005. http://www.dsls.usra.edu/meetings/bio2005/ , Jan-2005
Abstracts for Journals and Proceedings	Chien C, Baker K. "Pose estimation for servicing of orbital replacement units in a Cluttered Environment." IEEE Conf. on Robotics and Automation, New Orleans, LA, April 2004. IEEE Conf. on Robotics and Automation, April 2004. , Apr-2004